

Data Science Capstone - Weekly Report 2

Project: MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

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Last Week Progress / Topics Discussed:

- Conducted a structured literature review across four thematic areas aligned to the revised project scope: Affective Computing Fundamentals, Wearable Physiological Sensing, Mood/Emotion Classification Models, and Accessible Feedback Design.
- Reviewed, vetted, and finalized 10 papers across these four sections, replacing the earlier AI-assisted preliminary search with manually selected and verified sources.
- Section 1 (Affective Computing Fundamentals): Picard's foundational text on affective computing, plus a paper on deep-seeded clustering for emotion recognition from wearable physiological sensors, grounded in Russell's circumplex model of valence and arousal.
- Section 2 (Wearable Physiological Sensing): a systematic review of emotion recognition from wearable physiological signals, a study using the Fitbit-based LifeSnaps dataset for anxiety prediction, and a feasibility study documenting real constraints of pulling Fitbit API data (e.g., HRV is only available during sleep, not as a live daytime signal).
- Section 3 (Mood/Emotion Classification Models): a systematic review confirming Random Forest and SVM as the dominant classical ML approaches for wearable-based stress detection, and a paper on lightweight, personalized on-device emotion classification models relevant to small-sample, mobile deployment.
- Section 4 (Accessible Feedback Design): the iFeeling system for vibrotactile rendering of emotions on mobile phones, a systematic literature review of haptic feedback for blind and low-vision users, and a qualitative study of mobile application accessibility among Singapore's visually impaired community.
- Drafted a full literature review write-up organized by the four sections above, plus a synthesis paragraph, with in-text citations in IEEE numbered style and a verified reference list.
- Identified the research gap the project addresses: no existing system integrates consumer-wearable sensing, mood classification, and accessible closed-loop feedback into a single mobile-first pipeline.

Plan for Next Week:

- Review the literature review draft with Prof. Zantout and incorporate feedback.
- Identify candidate physiological signals and feature sets (heart rate, HRV, motion) accessible via the Fitbit/Apple Watch APIs for the classification pipeline, based on Section 2 findings.
- Begin evaluating classifier approaches short-listed in Section 3 (Random Forest, lightweight/personalized models) for feasibility given the project's data constraints.

To-Do List:

- Finalize literature review formatting and citations (confirm author details for the iFeeling source).
- Share the literature review draft with the supervisor for review.
- Research Fitbit/Apple Watch developer APIs to determine accessible data fields ahead of sensing-layer design.